Exemplar-Free Lifelong Hyperspectral Image Classification With Spectral Consistency

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***Abstract*— Hyperspectral image (HSI) classification suffers from severe catastrophic forgetting in exemplar-free lifelong learning, where models must continuously learn new land-cover categories without accessing historical training samples. This challenge persists due to high-dimensional spectral-volumetric complexity and cross-task spectral drift, which current methods inadequately address. We propose HyperSC, a novel framework that synergizes spectral-consistent auxiliary sample synthesis with stability-plasticity fused learning. The framework consists of three key components: a spectral consistency model inversion (SCMI) module, a spectral progressive enhancement (SPE) module, and a fusion distillation learning (FDL) module. The SCMI module synthesizes class-conditional auxiliary samples through spectral moment matching, in which the mean and variance of each spectral band are constrained to match class-specific real historical data distributions, thereby achieving spectral consistency. The SPE module injects class-specific Gaussian noise and applies momentum-based updating to enhance sample diversity while preserving spectral fidelity. The FDL module jointly trains on fused real and auxiliary samples by coordinating cross-entropy classification, output-layer knowledge distillation, and intermediate-layer feature alignment, thereby enabling plasticity for new class learning while maintaining stability against catastrophic forgetting for previous tasks. Extensive experiments on three HSI datasets (Indian Pines, Houston, and Salinas) demonstrate HyperSC's superiority compared to previous exemplar-free lifelong learning methods. The code is available at https://github.com/lzlsxs/hypersc**

***Index Terms*— Hyperspectral image (HSI) classification, incremental learning, lifelong learning, model inversion, spectral consistency.**

I. Introduction

HYPERSPECTRAL images (HSIs) possess unique and rich high-dimensional spectral information, encompassing tens to hundreds of contiguous spectral bands across a broad range from visible to shortwave infrared light. As a pixel-level classification task, HSI classification holds significant application value in various fields such as environmental monitoring [1], mineral exploration [2], military target reconnaissance [3], and smart agriculture [4].

Although deep learning-based HSI classification has achieved remarkable progress [5], [6], [7], [8], current methodologies rely on static and comprehensive training sets for offline training, which encompass all classes to be classified. However, in practical applications, due to the continuous increase in land-cover categories, models are required to continually acquire new knowledge and adapt to new classification tasks. This leads to catastrophic forgetting [9], [10], where the model forgets previously acquired knowledge while acquiring new knowledge. To address this challenge, the concept of lifelong learning (also known as continual learning or incremental learning) has emerged, aiming to enable computer vision systems to continuously acquire new classification abilities for new tasks while retaining their classification abilities for previous tasks [11], [12], [13]. Lifelong learning can be categorized into task incremental learning (TIL), class incremental learning (CIL), and domain incremental learning (DIL) based on learning scenarios [14], [15]. This study focuses on CIL, and all subsequent references to lifelong learning refer to CIL.

Exemplar-replay methods are considered a highly effective strategy for lifelong learning [16], [17], [18], [19], [20]. They select representative samples or train generators for each class of previous tasks, storing them for replay in subsequent tasks. Recently, several exemplar-replay methods tailored for HSI classification have been proposed [21], [22], [23], [24], [25], yielding certain results. However, exemplar-replay methods face challenges in practical applications, such as data privacy concerns and limited storage capacity. Consequently, lifelong learning must address the more challenging scenario of exemplar-free lifelong learning. In this scenario, the model cannot access any training data or generators from previous tasks when learning a new task.

Current approaches for exemplar-free lifelong learning comprise five main paradigms: distillation-based methods [15], [26], [27], [28], [29]; prototype-based methods [30], [31], [32]; ViT-based methods [33], [34], [35], [36], [37]; multimodal methods [38], [39], [40], [41]; and synthetic sample-based methods (primarily model inversion methods) [42], [43], [44], [45], [46]. Distillation-based methods mitigate catastrophic forgetting via dual mechanisms: a distillation loss preserves prior knowledge, while a cross-entropy loss acquires new knowledge. However, exclusive reliance on knowledge

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distillation leads to suboptimal performance [44], [45]. Prototype-based methods learn and store a prototype for each class by calculating the mean of the feature map from the penultimate layer across all samples in that class. In subsequent tasks, the model employs the nearest mean of exemplars (NMEs) method for classification, which calculates the distance between the new sample and each prototype, and selects the class with the smallest distance as the prediction result. However, with the continuous learning of new tasks, the prototypes of old classes inevitably undergo feature drift and cannot effectively represent the discriminative features of their corresponding classes [30], [31], [32]. Vision Transformer (ViT)-based methods start from a pretrained ViT and fine-tune it through methods such as prompt learning to achieve lifelong learning. Since the pretrained ViT model already contains a large amount of prior knowledge, it can adapt to new tasks more quickly while maintaining the knowledge of old tasks. Different modalities can provide complementary information to each other [47], [48]. Multimodal methods utilize large models such as MiniGPT-4 [49], Llama [50], or CLIP [51], combining visual and language information to achieve rapid adaptation to new tasks while retaining knowledge of old tasks. However, both ViT-based and multimodal methods require pretrained models and have high computational and storage requirements. Synthetic sample-based methods (primarily model inversion methods) generate pseudo-samples for old tasks by inverting only the stored data distribution information of the previous model or using other strategies. The advantage of synthetic sample-based methods lies in the elimination of the need to store synthetic samples and generators that are discarded upon completion of each task. Additionally, the integration of few-shot learning [52], [53], [54] with lifelong learning will also be a direction for future research.

However, due to the high-dimensional spectral bands of HSIs, these methods encounter the following challenges in exemplar-free lifelong learning for HSI classification.

1) Current pretrained ViTs and multimodal foundation models (e.g., CLIP) are predominantly optimized for RGB imagery. For HSI classification tasks, there is a lack of a unified pretrained ViT model or large-scale multimodal model for lifelong learning. Additionally, adapting these models incurs prohibitive computational and storage costs.
2) The effectiveness of solely relying on knowledge distillation in mitigating catastrophic forgetting is limited [44], [45]. Meanwhile, prototype-based methods inevitably encounter the issue of feature drift [30], [32]. This hinders their direct and effective application in lifelong learning for HSI classification.
3) Due to the high-dimensional spectral bands of HSI, existing synthetic sample-based methods struggle to generate high-quality synthetic samples that have spectral distributions consistent with real samples for lifelong learning.

These challenges have led to relatively limited research on exemplar-free lifelong learning for HSI classification. To address these challenges, we propose HyperSC, a novel exemplar-free lifelong learning framework for HSI classification based on spectral consistency. This framework consists of a spectral consistency model inversion (SCMI) module, a spectral progressive enhancement (SPE) module, and a fusion distillation learning (FDL) module. In HSIs, despite exhibiting similar data distributions to real samples, synthetic samples generated through model inversion are termed auxiliary samples due to some differences in visual and other attributes compared to real samples. The main contributions of this article are summarized as follows.

1) To address the challenge of generating high-quality HSI samples caused by high-dimensional spectral bands, we propose the SCMI module. This module synthesizes class-conditional auxiliary samples through spectral moment matching, where the mean and variance of each spectral band are constrained to align with class-specific historical data distributions, thereby achieving spectral consistency.
2) In HyperSC, the proposed SPE module employs class-specific Gaussian noise injection with momentum updating, enhancing sample diversity while preserving spectral fidelity.
3) The FDL module jointly trains on fused real and auxiliary samples by coordinating cross-entropy classification, output-layer knowledge distillation, and intermediate-layer feature alignment, thereby enabling plasticity for new class learning while maintaining stability against catastrophic forgetting for previous tasks.

We conduct extensive experiments on three HSI datasets, demonstrating that our proposed exemplar-free lifelong learning framework, HyperSC, effectively mitigates catastrophic forgetting and continuously acquires new knowledge, outperforming previous exemplar-free lifelong learning methods. The subsequent sections of this article are organized as follows. Section II introduces related research work on model inversion and knowledge distillation. Section III details our proposed HyperSC. Section IV describes the datasets, experimental settings, experimental results, and analysis. Section V summarizes the work in this study.

II. Related Work

A. Model Inversion

In contrast to data-driven generative paradigms such as diffusion models [55], model inversion enables the extraction of information about the original training data distribution from the already provided classification network without accessing the original training data and utilizes this information to generate or reconstruct synthetic images that are similar to the original training data [43]. Mahendran and Vedaldi [56] aimed to explore deep image representations by inverting the information of the image representation and utilizing generic natural image priors to optimize a random noise image, thereby achieving image reconstruction. Mordvintsev et al. [57] proposed DeepDream, which inputs a random noise image into a pretrained network. In each iteration, the gradient is calculated based on the activation values of a given layer, and the pixel values of the noise image are updated in the direction of gradient ascent instead

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of updating the network weights. This process maximizes the activation values of the given layer, ultimately achieving image reconstruction. Yin et al. [42] proposed DeepInversion, which introduces an additional regularization term based on DeepDream. By optimizing this noise image to ensure that its feature statistics match those of the original training data, DeepInversion can be applied in scenarios such as data-free model compression and pruning, data-free knowledge distillation and transfer, and data-free lifelong learning. Previous model inversion methods typically relied on optimizing random noise images, requiring numerous iterations for each image generation and resulting in low computational efficiency. To address this issue and accommodate lifelong learning scenarios, Smith et al. [43] proposed the always be dreaming (ABD) method, applying model inversion to lifelong learning. ABD optimizes a generator without the training samples, rather than a random noise image for model inversion, thereby significantly improving computational efficiency and providing a new solution for lifelong learning. Gao et al. [44] and Qiu et al. [45] proposed the R-DFCIL and DCMI methods to address the domain shift issue between synthetic samples generated through model inversion and real samples in lifelong learning. Nevertheless, due to the high-dimensional spectral bands of HSIs, there has been limited attention on lifelong learning for HSI classification based on model inversion.

B. Knowledge Distillation

Knowledge distillation, proposed by Hinton et al. [58], is an efficient model compression technique aimed at transferring knowledge from complex, large teacher models to simpler, lightweight student models. Knowledge distillation can be categorized into three main types: logit-based [58], [59], [60], feature-based [61], [62], [63], and relation-based knowledge distillation [64], [65], [66]. Logit-based knowledge distillation enables the student model to mimic the prediction logit distribution of the teacher model, thereby ensuring a high similarity in their prediction abilities. Feature-based knowledge distillation focuses on the consistency of the intermediate representations in the hidden layers of the student model with those of the teacher model, by mimicking the teacher's hidden layer features to achieve knowledge distillation. Relation-based knowledge distillation aims to transfer the structural relationships between samples or contexts in the teacher model to the student model. In recent years, data-free knowledge distillation methods have garnered significant attention [67], [68], [69], [70], [71]. These methods adhere to the inversion-and-distillation paradigm. In the inversion process, synthetic samples that are consistent with the data distribution of the real samples are generated using the information learned from the teacher model. In the distillation phase, the student model is trained to match the prediction logit distribution or intermediate feature distribution of the teacher model on these synthetic samples. This approach offers notable advantages in protecting data privacy, mitigating data dependency, and reducing storage constraints.

In the field of lifelong learning, knowledge distillation also plays a crucial role. LWF [26] applies logit-based knowledge distillation to lifelong learning, effectively transferring knowledge learned from previous models to the current model. LFL [27] further preserves previously learned knowledge by aligning the intermediate representations of the last hidden layer between the previous and current models via feature-based knowledge distillation. As research on lifelong learning progresses, knowledge distillation-based lifelong learning methods have attracted increasing attention and study [15], [28], [29]. HSIs contain rich spectral–spatial information [72]. In recent years, knowledge distillation-based lifelong learning has also been introduced to HSI classification, some methods tailored to the unique characteristics of HSIs have been proposed [21], [22], [23], [24], [25]. However, these methods primarily rely on exemplar-replay.

III. Proposed Method

In this section, we provide a detailed introduction to our proposed HyperSC framework. Section III-A introduces the problem setup for exemplar-free lifelong learning. Section III-B provides an overview of the proposed framework. Section III-C describes the SCMI module. Section III-D introduces the SPE module. Section III-E describes the FDL module.

A. Problem Setup

In exemplar-free lifelong learning, models sequentially acquire knowledge from a series of tasks $\mathcal{D} = \{\mathcal{D}_1, \ldots, \mathcal{D}_T\}$. The dataset for each task $\mathcal{D}_t = \{X_t, Y_t\}$ consists of the input samples $X_t = \{x_i\}_{i=1}^{n_t}$ and their corresponding labels $Y_t = \{y_i \in \mathcal{C}_t\}_{i=1}^{n_t}$, where $\mathcal{C}_t$ represents the set of classes, $|\mathcal{C}_t|$ is the number of classes, and n_t denotes the number of all samples in task t. The classes in each task do not overlap with those in other tasks, that is, $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset$ if $i \neq j$. Throughout the learning process, the model is prohibited from storing training samples or generators. When a model learns a new task t, it can only access training data $\mathcal{D}_t = \{X_t, Y_t\}$ from the current task. However, the evaluation criteria for assessing its performance must encompass all classes $\cup_{i=1}^{t}\mathcal{C}_i$ from task 1 to t. This limitation often results in catastrophic forgetting.

The goal of lifelong learning is to construct a classification network that can correctly classify all seen classes $\cup_{i=1}^{t}\mathcal{C}_i$ from task 1 to t after learning task t.

Notably, in HSIs, although synthetic samples generated through model inversion exhibit data distributions similar to those of real samples, they are termed auxiliary samples because of some differences in visual and other attributes compared to real samples.

B. Overview of the Proposed Framework

Our proposed exemplar-free lifelong learning framework, HyperSC, consists of three main components: the SCMI module, the SPE module, and the fusion distillation learning (FDL) module. The overview of the proposed framework is presented in Fig. 1. Each task consists of three stages.

1) *Auxiliary Sample Generation:* In this phase, we utilize the SCMI module to invert the previously trained classification network, generating a batch of class-conditional

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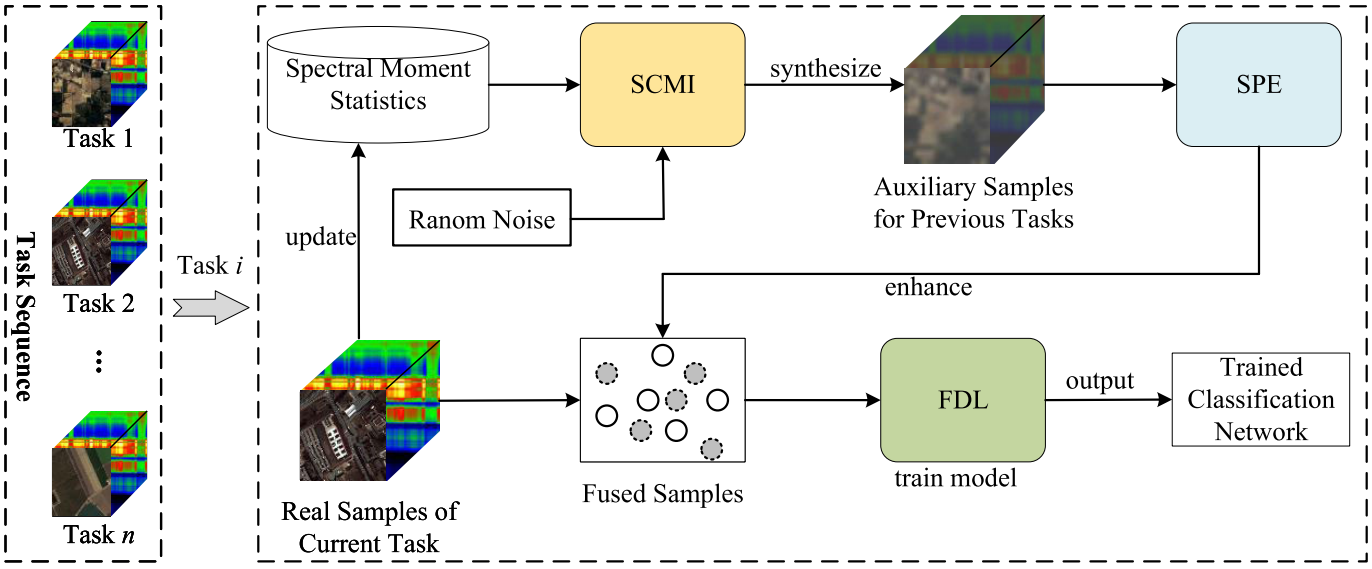


Fig. 1. Flowchart of the proposed HyperSC framework. The SCMI module synthesizes auxiliary samples by optimizing random noise, leveraging only preserved spectral moment statistics from previous tasks. The SPE module enhances sample diversity while maintaining spectral fidelity through Gaussian noise injection. The FDL module jointly trains on fused real and auxiliary samples, enabling the simultaneous acquisition of new classification abilities while retaining prior knowledge. Critically, all training samples (both real and auxiliary) are discarded post-task, with only updated spectral moment statistics retained for subsequent tasks.

auxiliary samples by optimizing random noise, leveraging only the preserved spectral moment statistics from previous tasks. To minimize computational overhead, we generate only a single batch of auxiliary samples encompassing all classes from previous tasks, which are reused to support subsequent classification network training. These auxiliary samples are discarded immediately after the completion of the current task.

2) *SPE:* The SPE module further enhances these auxiliary samples by progressively adding random class-specific noise based on the statistical distribution of real samples. This noise addition enhances sample diversity while maintaining spectral fidelity, ensuring that these auxiliary samples are as close as possible to the data distribution of the real samples.
3) *FDL:* The FDL module integrates real samples of the current classes with auxiliary samples of previous classes. Using the respective characteristics of these samples, the FDL module learns new classification abilities for the current task while retaining the classification abilities acquired from previous tasks. Notably, after each task, we discard these auxiliary samples and real samples.

C. Spectral Consistency Model Inversion

HSIs, characterized by their high-dimensional spectral bands, pose challenges in generating high-quality auxiliary samples whose spectral distributions closely match real samples via model inversion. Consequently, such limitations impact the performance of subsequent tasks, including classification. To address these challenges, we propose the SCMI module. Fig. 2 shows the illustrative diagram of auxiliary samples generation by the SCMI module. The SCMI module synthesizes class-conditional auxiliary samples by inverting the previously trained classification network, without accessing any training data. Specifically, it initializes random noise

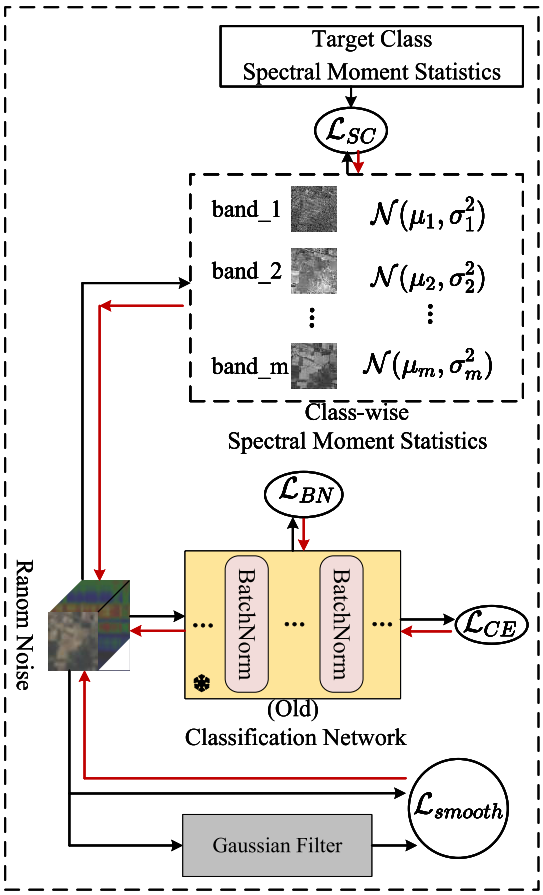


Fig. 2. SCMI module synthesizes class-conditional auxiliary samples via spectral moment matching (μ and σ^2), and multiconstraint coordination (cross-entropy loss, BN distribution alignment, and local spatial smoothness loss), achieving spectral consistency. Black arrows represent forward propagation, while red arrows represent backpropagation.

and iteratively optimizes it under spectral moment constraints (mean and variance per spectral band) guided by class-specific statistics, thereby aligning auxiliary samples with the real

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data distributions of target classes. The implementation details are as follows: for each task, we compute the mean and variance of all real samples for every new class across all spectral bands. During auxiliary sample generation, spectral moment statistics (mean and variance) are computed for each spectral band of the generated auxiliary samples. These measurements are subsequently compared with precalculated spectral moments derived from real samples of corresponding classes in prior tasks. The distributional discrepancy is quantified using the Kullback–Leibler (KL) divergence under the Gaussian assumption, thereby enhancing the quality of auxiliary samples. It can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{SC}}(\hat{x}) &= \frac{1}{B}\sum_{b=1}^{B}\mathrm{KL}\left(\mathcal{N}\left(\mu_{\hat{y}b},\sigma_{\hat{y}b}^2\right)\|\mathcal{N}\left(E\left(\hat{x}_b\right),\sigma^2\left(\hat{x}_b\right)\right)\right)\\ &= \log\frac{\sigma\left(\hat{x}_b\right)}{\sigma_{\hat{y}b}} - \frac{1}{2}\left(1-\frac{\sigma_{\hat{y}b}^2+\left(\mu_{\hat{y}b}-E\left(\hat{x}_b\right)\right)^2}{\sigma^2\left(\hat{x}_b\right)}\right)\end{aligned} \tag{1}$$

where $\hat{x}$ represents auxiliary samples being optimized, $\hat{y}$ represents the classification labels of the auxiliary samples, $\mu_{\hat{y}b}$ and $\sigma_{\hat{y}b}$, respectively, represent the mean and standard deviation of the real training data labeled as $\hat{y}$ on the bth spectral band, $E(\hat{x}_b)$ and $\sigma(\hat{x}_b)$, respectively, represent the mean and standard deviation of at the bth spectral band for auxiliary samples, and B represents the total number of spectral bands.

Additionally, the SCMI module integrates cross-entropy loss, local smoothness loss [43], and batch normalization (BN) loss [43], [68], collectively forming a final loss function for auxiliary sample generation. The objective of this final loss function is to ensure that the generated auxiliary samples maintain a high degree of consistency with the real samples in terms of quality, reliability, local smoothness, and especially in the distribution of spectral bands.

1) Cross-Entropy Loss: The cross-entropy loss measures the discrepancy between model predictions and ground-truth labels. To enhance the quality of auxiliary samples, we introduce a temperature scaling mechanism that softens the output probability distribution. Specifically, a large temperature value ($T > 1$) flattens prediction logits, encouraging generated samples to align with class-discriminative features and improving their representativeness for downstream tasks. The loss is formulated as

$$\mathcal{L}_{\mathrm{CE}}\left(\phi\left(\hat{x}\right),\hat{y}\right) = -\sum_{i=1}^{S}\hat{y}_i\log\left(\frac{\exp\left(\phi\left(\hat{x}\right)_i/T\right)}{\sum_{j=1}^{S}\exp\left(\phi\left(\hat{x}\right)_j/T\right)}\right) \tag{2}$$

where $\hat{x}$ represents auxiliary samples being optimized, $\hat{y}$ represents the classification labels of the auxiliary samples, ϕ denotes the frozen old classification model, T is the temperature scaling parameter, and S is the number of classes.

2) Local Smoothness Loss: HSIs inherently exhibit local spectral–spatial smoothness: neighboring pixels share similar spectral characteristics due to intrinsic spatial continuity. To leverage this spectral–spatial prior, we apply a Gaussian spatial filter to smooth the auxiliary samples, enforcing alignment with the intrinsic smoothness property of HSI data. The smoothness loss is quantified through ℓ_2-norm minimization between the original and filtered samples as follows:

$$\mathcal{L}_{\mathrm{smooth}}\left(\hat{x},\hat{x}_{\mathrm{smooth}}\right) = \|\hat{x}-\hat{x}_{\mathrm{smooth}}\|_2^2 \tag{3}$$

where $\hat{x}$ denotes the auxiliary samples being optimized and $\hat{x}_{\mathrm{smooth}}$ represents the filtered version of the auxiliary samples obtained through the application of a Gaussian filter.

3) BN Statistics Loss: Haroush et al. [68] introduced BN, a technique designed to enhance the training speed and stability of models. During the training phase, BN calculates the moving mean as well as the moving variance for each channel feature across all batches within the layer. These moving mean and moving variance are then used during inference to normalize the features of each BN layer channel. When training for the generation of auxiliary samples, the moving mean and moving variance obtained from a previously trained classification model can serve as target values for each BN layer. The random noise is fed into the classification model to compute the mean and variance at each BN layer. It is then iteratively optimized by minimizing the KL divergence between the computed statistics and the previously obtained target BN statistics. It can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{BN}}(\hat{x}) &= \frac{1}{\mathrm{L}}\sum_{l=1}^{\mathrm{L}}\mathrm{KL}\left(\mathcal{N}\left(\mu_l,\sigma_l^2\right)\|\mathcal{N}\left(\hat{\mu}_l(\hat{x}),\hat{\sigma}_l^2(\hat{x})\right)\right)\\ &= \log\frac{\hat{\sigma}_l(\hat{x})}{\sigma_l} - \frac{1}{2}\left(1-\frac{\sigma_l^2+\left(\mu_l-\hat{\mu}_l(\hat{x})\right)^2}{\hat{\sigma}_l^2(\hat{x})}\right)\end{aligned} \tag{4}$$

where $\hat{\mu}_l(\hat{x})$ and $\hat{\sigma}_l^2(\hat{x})$) represent the mean and variance of the features at the lth BN layer for a given batch of auxiliary samples, respectively, μ_l and σ_l^2 represent the moving mean and moving variance statistics of the features at the lth BN layer previously collected during the training process of the classification model, respectively, and L represents the total number of BN layers.

The final loss function for generating auxiliary samples can be formulated as follows:

$$\begin{aligned}\mathcal{L}_{\mathrm{final}} &= \mathcal{L}_{\mathrm{CE}}\left(\phi\left(\hat{x}\right),\hat{y}\right)) + \lambda_1\mathcal{L}_{\mathrm{SC}}\left(\hat{x}\right)\\ &\quad + \lambda_2\mathcal{L}_{\mathrm{smooth}}\left(\hat{x},\hat{x}_{\mathrm{smooth}}\right) + \lambda_3\mathcal{L}_{\mathrm{BN}}\left(\hat{x}\right).\end{aligned} \tag{5}$$

The optimization of random noise to generate auxiliary samples via the SCMI module is presented in Algorithm 1.

D. Spectral Progressive Enhancement

While SCMI-synthesized auxiliary samples exhibit partial spectral distribution consistency with historical real data, significant discrepancies persist in visual quality and other characteristics, particularly critical in HSIs with 200+ spectral bands. To bridge this gap, we propose the SPE module with dual optimization objectives: spectral fidelity preservation and augmentation diversity, as shown in Fig. 3.

First, class-conditional Gaussian noise

$$\epsilon \sim \mathcal{N}\left(\mu_c,\sigma_c^2\right) \tag{6}$$

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Algorithm 1 Spectral Consistency Model Inversion

Input: θ_{old} //old feature extractors
Input: ϕ_{old} //old classifiers
$\hat{x} \sim \mathcal{N}(0, I)$ //Initialize auxiliary sample
for $n = 1$ to $Epoch$ **do**
  $\hat{f}_{old} \leftarrow \theta_{old}(\hat{x})$ //feature from old network
  $\hat{p}_{old} \leftarrow \phi_{old}(\hat{f}_{old})$ //old network prediction
  $\mathcal{L}_{CE}(\hat{y}, \hat{p}_{old}/T)$ ←Calculate the loss with (2)
  $\hat{x}_{smooth}$ ←Process with Gaussian filter
  $\mathcal{L}_{smooth}(\hat{x}, \hat{x}_{smooth})$ ←Calculate the loss with (3)
  $\mathcal{L}_{BN}(\hat{x})$ ←Calculate the loss with (4)
  $\mathcal{L}_{\mathrm{SC}}(\hat{x})$ ←Calculate the loss with (1)
  $\mathcal{L}_{total} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{\mathrm{SC}} + \lambda_2 \mathcal{L}_{smooth} + \lambda_3 \mathcal{L}_{BN}$
  $\hat{x} \leftarrow \hat{x} - \eta \nabla_{\hat{x}}(\mathcal{L}_{total})$ //Update auxiliary sample
end for
Output: $\hat{x}$

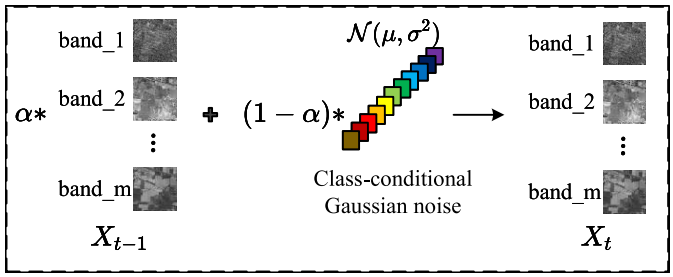


Fig. 3. SPE module enhances sample diversity via class-specific Gaussian noise injection and momentum updating, dynamically refining auxiliary samples while preserving spectral fidelity. t denotes the tth training iteration.

is injected into the auxiliary samples. Here, $\boldsymbol{\mu}_c \in \mathbb{R}^B$ and $\boldsymbol{\sigma}_c \in \mathbb{R}^B$ denote the class-specific mean vector and the standard deviation vector computed from real samples of the category c, respectively. This process injects authentic spectral noise while preserving distributional alignment. Notably, both μ and σ align with the dimensions of the spectral band of the samples, ensuring that each spectral band undergoes independent enhancement.

Second, we implement a momentum-based updating strategy formulated as

$$\hat{\boldsymbol{x}}_t = \alpha \hat{\boldsymbol{x}}_{t-1} + (1-\alpha)\epsilon \tag{7}$$

where α is the momentum coefficient and t denotes the tth training iteration. This mechanism ensures augmentation diversity through progressive noise injection, while strictly preserving spectral fidelity via momentum-based updates.

E. Fusion Distillation Learning

In the phase of training the current classification model, we can acquire real samples from the current task and auxiliary samples corresponding to previous tasks. Given the limited quantity and inherent distributional discrepancies of auxiliary samples generated for previous tasks, we face the challenge of effectively integrating these auxiliary samples with real samples to train the current classification model. To address this, we propose the FDL module, which leverages the abundance of real samples and the scarcity of auxiliary samples to jointly optimize the classification model, enabling simultaneous learning for both current and previous classes. Specifically, during each training batch, we fuse auxiliary samples with the current batch of real samples for joint training. The SPE module is applied in this fusion process to inject diversity into auxiliary samples across batches, thus mitigating overfitting. We employ the cross-entropy loss function to simultaneously optimize the classification model for current and previous classes. Additionally, we introduce prediction logit distillation and spectral–spatial feature distillation to transfer the feature representation and classification abilities from the previously trained model. Consequently, the classifier maintains stable performance in continuously updated tasks, facilitating the continuous accumulation and transfer of knowledge. Fig. 4 illustrates the overall framework of the FDL module. The optimization objective for the FDL module consists of the following three components.

1) Fusion Learning: For the fused real and auxiliary samples, the component employs a cross-entropy loss function to quantify the divergence between the model predictions and the ground-truth labels. This enables the model to jointly optimize its discriminative capabilities for current and previous classes. The loss function ensures that the model adapts to the class distributions of old and new tasks, thus preserving globally consistent decision boundaries during the fusion training process. It can be formulated as follows:

$$\mathcal{L}_{\mathrm{FL}}(\phi(x), y) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{S} y_i^c \log\left(\frac{\exp\left(\phi(x_i)^c\right)}{\sum_{j=1}^{S}\exp\left(\phi(x_i)^j\right)}\right) \tag{8}$$

where $x = [x_{\mathrm{real}}, x_{\mathrm{aux}}]$ denotes fused real and auxiliary samples, y indicates the labels of the fused samples, ϕ represents the current classification model, S indicates the total number of classes encountered so far, and N represents the number of samples in this batch.

2) Fusion Knowledge Distillation: The distillation process feeds the fused samples into both the current classification model and the previously trained model, minimizing the cross-entropy between their prediction distributions. This alignment enforces consistency in discriminative abilities for all classes of previous tasks between the two models, thereby facilitating the transfer of classification knowledge from the old model to the new one. Specifically, the cross-entropy constraint encourages the output distribution of the current model to approximate that of the previous model, ensuring that the current model retains the classification abilities of old tasks while learning new tasks

$$\mathcal{L}_{\mathrm{FKD}}\left(\phi^t(x), \phi^{t-1}(x)\right) = -\frac{1}{S}\sum_{c=1}^{S} p_c^{t-1}\log\left(p_c^t\right) \tag{9}$$

$$p_c = \frac{\exp\left(\phi(x)_c/T\right)}{\sum_{i=1}^{S}\exp\left(\phi(x)_i/T\right)} \tag{10}$$

where $x = [x_{\mathrm{real}}, x_{\mathrm{aux}}]$ denotes the fused real and auxiliary samples, ϕ represents the classification model, S represents the total number of classes up to the previous task, t and $t-1$ denote the current task and the previous task, respectively, and T is the temperature scaling parameter.

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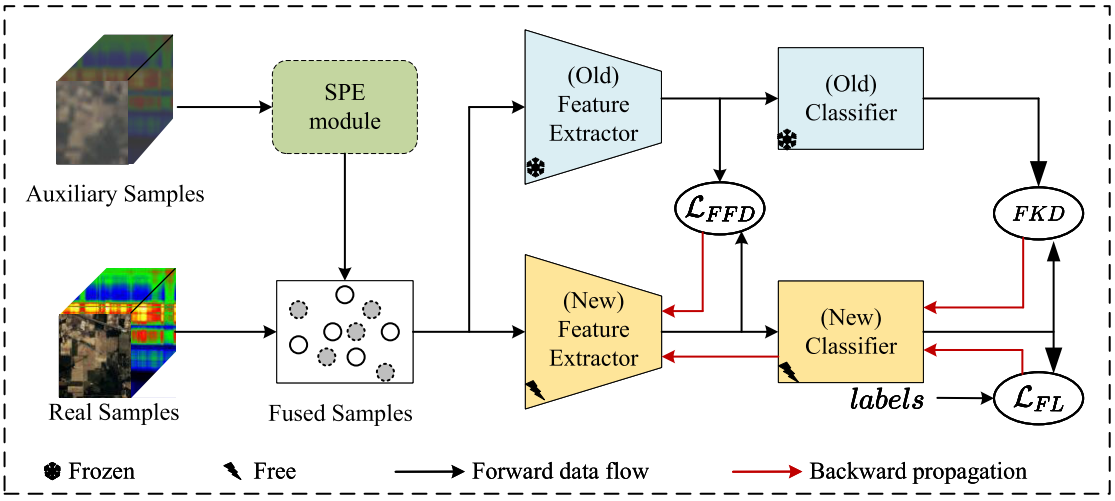


Fig. 4. FDL module integrates real and auxiliary samples, optimized via cross-entropy loss and dual-distillation (prediction logit + spectral–spatial feature) to ensure stable performance and knowledge transfer in lifelong learning.

3) Fusion Feature Distillation: This component employs a feature-layer distillation strategy by feeding the fused samples into both the current model and the previous model. It computes the L2-norm of the outputs from the last feature layer of both models to construct a distillation loss function. This enforces the current model to learn and preserve the feature representation abilities acquired by the previous model. Specifically, minimizing the L2-norm between the feature-layer outputs of the two models ensures the continuity of feature representation in lifelong learning. It can be formulated as follows:

$$\mathcal{L}_{\text{FFD}}\left(\phi^t(x), \phi^{t-1}(x)\right) = \frac{1}{N}\sum_{i=1}^{N} \|\theta^{t-1}(x_i) - \theta^t(x_i)\|_2^2 \quad (11)$$

where $x = [x_{\text{real}}, x_{\text{aux}}]$ denotes the fused real and auxiliary samples, θ represents the feature extractor, t and $t-1$ denote the current task and the previous task, respectively, and N represents the number of samples in this batch.

The final loss function for training the current classification model can be formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FL}}(\phi(x), y) + \lambda_1 \mathcal{L}_{\text{FKD}}\left(\phi^t(x), \phi^{t-1}(x)\right) + \lambda_2 \mathcal{L}_{\text{FFD}}\left(\theta^t(x), \theta^{t-1}(x)\right). \quad (12)$$

Detailed training steps of the FDL are summarized in Algorithm 2.

Algorithm 2 Fusion Distillation Learning

Input: $D_t = \{X_t, Y_t\}$ //current task samples
Input: $\hat{D}_t = \{\hat{X}_t, \hat{Y}_t\}$ //auxiliary samples
Input: $\theta_{old}, \theta_{new}$ //old and new feature extractors
Input: ϕ_{old}, ϕ_{new} //old and new classifiers
Training new feature extractor and classifier
for $n = 1$ to $Epoch$ **do**
  for $(x_t, y_t) \in D_t$ **do**
    $x_{com} = x_t \cup \hat{x}_t, \quad y_{com} = y_t \cup \hat{y}_t$ //fused samples
    $f_{old} \leftarrow \theta_{old}(x_{com})$ //feature from old network
    $p_{old} \leftarrow \phi_{old}(f_{old})$ //old network prediction
    $f_{new} \leftarrow \theta_{new}(x_{com})$ //feature from new network
    $p_{new} \leftarrow \phi_{new}(f_{new})$ //new network prediction
    $\mathcal{L}_{FL}(p_{new}, y_{com}) \leftarrow$ Calculate the loss with (8)
    $\mathcal{L}_{FKD}(p_{old}, p_{new}) \leftarrow$ Calculate the loss with (9)
    $\mathcal{L}_{FFD}(f_{old}, f_{new}) \leftarrow$ Calculate the loss with (11)
    $\mathcal{L}_{total} = \mathcal{L}_{FL} + \lambda_1 \mathcal{L}_{FKD} + \lambda_2 \mathcal{L}_{FFD}$
    $\theta, \phi \leftarrow \theta, \phi - \nabla_{\theta,\phi}(\mathcal{L}_{total})$
  end for
end for
$\hat{x} \leftarrow \emptyset, \hat{y} \leftarrow \emptyset$ //discard auxiliary samples
Output: θ, ϕ

IV. Experiments

A. Hyperspectral Datasets

In this study, three HSI datasets are used for the experiments. The details of each dataset are described below.

1) Indian Pines: The Indian Pines dataset was collected in 1992 using the airborne visible–infrared imaging spectrometer (AVIRIS) over a pine forest in Indiana, USA. It consists of 145 × 145 pixels and 220 spectral bands, of which 20 bands are affected by water absorption, resulting in 200 practical bands for analysis. It includes 10 366 labeled pixels, categorized into 16 distinct classes.

2) Houston: The Houston dataset, collected by the National Center for Airborne Laser Mapping, covers the campus of the University of Houston. It is a crucial resource for remote sensing technology and land-cover classification research. The dataset comprises 349 × 1905 pixels, with each pixel having 144 spectral bands. A total of 15 029 pixels are labeled, distinguishing 15 different land-cover types.

3) Salinas: The Salinas dataset, acquired by NASA's AVIRIS over agricultural lands in California's Salinas Valley, contains spectral signatures of diverse crops and vegetation types. Comprising 512 × 217 spatial pixels with 224 spectral bands spanning 400–2500 nm, the dataset retains 204 calibrated bands after eliminating 20 channels affected by water vapor absorption. It provides 16 annotated land-cover

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TABLE I
NUMBER OF TRAINING AND TESTING SAMPLES PER CLASS FOR THE INDIAN PINES, HOUSTON, AND SALINAS DATASETS

Indian Pines			Houston			Salinas		
Class Name	Train	Test	Class Name	Train	Test	Class Name	Train	Test
Corn Mintill	50	784	Healthy Grass	198	1053	Brocoli Green Weeds1	200	1809
Corn	50	184	Stressed Grass	190	1064	Brocoli Green Weeds2	200	3526
Grass Pasture	50	447	Synthetic Grass	192	505	Fallow	200	1776
Grass Tree	50	697	Tree	188	1056	Fallow Rough Plow	200	1194
Hay Windrowed	50	439	Soil	186	1056	Fallow Smooth	200	2478
Soybean clean	50	564	Water	182	143	Stubble	200	3759
Wheat	50	162	Residential	196	1072	Celery	200	3379
Build Grass Tree Drives	50	330	Commercial	191	1053	Grapes Untrained	200	11071
Stone Steel Towers	50	45	Road	193	1059	Soil Vinyard Develop	200	6003
Alfalfa	15	39	Highway	191	1036	Corn Senesced Green Weeds	200	3078
Grass Pasture Mowed	15	11	Railway	181	1054	Lettuce Romaine 4wk	200	868
Oats	15	5	Parking Lot1	192	1041	Lettuce Romaine 5wk	200	1727
Corn Notill	50	1384	Parking Lot2	184	285	Lettuce Romaine 6wk	200	716
Soybean Notill	50	918	Tennis Court	181	247	Lettuce Romaine 7wk	200	870
Soybean Mintill	50	2418	Running Track	187	473	Vinyard Untrained	200	7068
Woods	50	1244				Vinyard Vertical Trellis	200	1607
Total	695	9671	Total	2832	12197	Total	3200	50929

categories with approximately 54 000 labeled pixels, serving as a critical benchmark for HSI classification and crop species discrimination.

We divide the training and testing sets for these three HSI datasets, with the class names and the corresponding number of training and testing samples for each class shown in Table I.

B. Experimental Setting

1) Environmental Setting: The experiments presented in this article are conducted using the following hardware and software configuration, which comprises an NVIDIA A16 GPU equipped with 15GB of VRAM, an Intel Xeon Gold 5318Y CPU, 14GB RAM, the Ubuntu 20.04 Operating System, Python 3.10 as the programming language, PyTorch 2.1.1 as the deep learning framework, CUDA version 11.8, and cuDNN version 8.

2) Parameter Setting: Initially, we uniformly reduced the spectral bands to 36 using principal component analysis (PCA) for preprocessing. Subsequently, sampling is performed with a patch size of 7, generating $7 \times 7 \times 36$ patches for classification. In the SPE module, the momentum coefficient is set to 0.5. The coefficients λ_1, λ_2, and λ_3 in (5) are uniformly set to 50, 0.001, and 50, respectively, across all datasets, whereas in (12), λ_2 maintains 0.1 universally with λ_1 configured as 10 for Indian Pines, 1 for Houston, and 0.01 for Salinas. For HSIs, we choose LSGA [7] as the backbone network. During the auxiliary sample generation, the Adam optimizer is used with a learning rate of 1×10^{-3}. For classifier training, stochastic gradient descent (SGD) is employed as the optimizer, with 100 training epochs, a learning rate of 0.05, and a batch size of 128.

3) Task Setting: Under the lifelong learning paradigm, each HSI dataset is partitioned into three incremental learning tasks. For the Indian Pines and Salinas datasets, following the class order specified in Table I, Tasks 1, 2, and 3 are assigned 6, 6, and 4 distinct classes, respectively. The Houston dataset is similarly divided into three tasks, each containing 5 classes based on the predefined sequence in Table I. During training, the model only accesses training samples from the current task. For evaluation, a cumulative testing strategy is employed: model performance is assessed on test data encompassing all previously learned tasks, thereby validating its knowledge retention ability throughout the lifelong learning process.

4) Evaluation Metrics: In the evaluation system for classification tasks, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient are used as performance metrics. For lifelong learning, OA is adopted to evaluate the model's classification performance across all learned classes upon the completion of each task. After all tasks are completed, the average OA across all tasks is calculated as the average task accuracy (Ave). Specifically, this study analyzes two key metrics: 1) the last task accuracy (LAC) for Task 3, which quantifies the model's cumulative knowledge retention throughout the lifelong learning process and 2) the Ave, which evaluates the model's stability and generalization capability across all tasks.

C. Parameter Sensitivity Analysis

1) Patch Size: To determine the optimal patch size, we conduct experiments on several different patch sizes, including 3×3, 5×5, 7×7, 9×9, 11×11, and 13×13. As illustrated in Fig. 5, in the Indian Pines dataset, the Ave peaks at 7×7 with a performance of 94.08%, maintaining relatively stable performance up to 9×9, followed by a slight decline thereafter. The LAC shows an increasing trend with larger patch sizes, stabilizing at 7×7 and continuing until 9×9. In the Salinas dataset, both LAC and Ave increase with patch size, reaching stability at 7×7 and remaining consistent until 11×11, after which there is a minor decrease. For the Houston dataset, both LAC and Ave maintain relative stability up to 9×9, followed by a noticeable decline beyond this point. In this study, we consistently set the patch size to 7×7.

2) Momentum Coefficient in SPE: We systematically investigate the impact of the momentum coefficient α in (7) through controlled experiments with $\alpha \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$, where smaller α values correlate with stronger noise injection intensity. As shown in Fig. 6, the model performance on the

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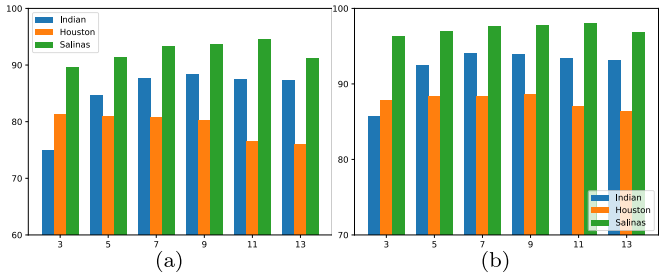


Fig. 5. Experimental results (%) of the different patch sizes on the Indian Pines, Houston, and Salinas datasets. (a) LAC. (b) Average task accuracy (Ave).

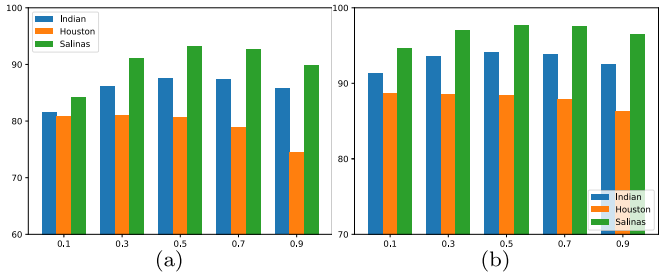


Fig. 6. Experimental results (%) of the different momentum coefficients (α) in (7) on the Indian Pines, Houston, and Salinas datasets. (a) LAC. (b) Average task accuracy (Ave).

TABLE II

ABLATION STUDY RESULTS (%) ON THE INDIAN PINES, HOUSTON, AND SALINAS DATASETS

Dataset	Setting	LAC (Task 3)	Ave
Indian	Baseline	56.44±0.71	57.04±0.22
	Baseline+SCMI	64.83±2.27	67.56±2.05
	Baseline+SCMI+SPE	87.18±0.98	90.62±0.28
	Baseline+SCMI+SPE+FDL	**87.62±1.34**	**94.08±0.66**
Houston	Baseline	22.90±0.47	52.81±0.42
	Baseline+SCMI	24.63±3.16	54.18±1.35
	Baseline+SCMI+SPE	80.34±3.54	88.20±1.22
	Baseline+SCMI+SPE+FDL	**80.75±1.07**	**88.43±0.80**
Salinas	Baseline	20.24±0.15	62.25±1.18
	Baseline+SCMI	27.91±6.81	69.93±3.08
	Baseline+SCMI+SPE	87.89±6.20	95.87±2.06
	Baseline+SCMI+SPE+FDL	**93.28±0.74**	**97.66±0.25**

Indian and Salinas datasets exhibits an unimodal distribution with increasing α, reaching its optimal value at $\alpha = 0.5$. For the Houston dataset, performance remains relatively stable when $\alpha \leq 0.5$, whereas exceeding this threshold shows a slight decline. These observations demonstrate a dual-aspect equilibrium: excessive noise degrades spectral fidelity while insufficient perturbation diminishes sample diversity. In this study, we uniformly set $\alpha = 0.5$.

D. Ablation Study

We conduct ablation experiments to assess the contribution of each module in our method to lifelong learning. The experimental results present the LAC and the Ave on the Indian Pines, Houston, and Salinas datasets, as shown in Table II, which are the average values after five runs with different random seeds. HyperSC without SCMI, SPE, and FDL is utilized as a baseline.

TABLE III

CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH CLASS ON THE INDIAN PINES DATASET

Class No.	Class Name	Task1 (6 classes)	Task2 (6 classes)	Task3 (4 classes)
0	Corn Mintill	99.52±0.53	99.21±0.31	84.69±2.55
1	Corn	100.00±0.00	98.26±0.94	97.17±0.63
2	Grass Pasture	97.18±0.52	86.71±2.85	78.30±2.90
3	Grass Tree	99.89±0.06	99.89±0.06	96.79±1.49
4	Hay Windrowed	100.00±0.00	99.91±0.11	99.86±0.11
5	Soybean Clean	94.08±0.95	89.43±2.65	75.71±5.23
6	Wheat		100.00±0.00	100.00±0.00
7	Build Grass Tree Drives		98.36±2.02	88.18±2.72
8	Stone Steel Towers		100.00±0.00	100.00±0.00
9	Alfalfa		90.26±1.92	94.36±1.92
10	Grass Pasture Mowed		100.00±0.00	100.00±0.00
11	Oats		100.00±0.00	100.00±0.00
12	Corn Notill			90.16±2.26
13	Soybean Notill			87.58±3.52
14	Soybean Mintill			80.53±3.60
15	Woods			95.72±0.89
	OA	98.38±0.20	96.26±0.83	87.62±1.34
	AA	98.44±0.19	96.83±0.59	91.82±0.73
	Kappa	97.99±0.25	95.63±0.97	85.84±1.51

In the Indian Pines dataset, adding the SCMI module to the baseline improves LAC from 56.44% to 64.83% (+8.39%) and Ave from 57.04% to 67.56% (+10.52%). With further addition of the SPE module, LAC increases to 87.18% (+22.35%) and Ave increases to 90.62% (+23.06%). Finally, after incorporating the FDL module, LAC reaches 87.62% (+0.44%), with Ave rising to 94.08% (+3.46%).

In the Houston dataset, after adding the SCMI module to the baseline, LAC increases from 22.90% to 24.63% (+1.73%) and Ave increases from 52.81% to 54.18% (+1.37%). Further addition of the SPE module leads to a significant improvement, with LAC increasing to 80.34% (+55.71%) and Ave improving to 88.20% (+34.02%). Lastly, after adding the FDL module, the LAC rises to 80.75% (+0.41%), while Ave improves to 88.43% (+0.23%).

In the Salinas dataset, after adding the SCMI module to the baseline, LAC increases from 20.24% to 27.91% (+7.67%) and Ave improves from 62.25% to 69.93% (+7.68%). The addition of the SPE module results in a substantial increase, with LAC rising to 87.89% (+59.98%) and Ave reaching 95.87% (+25.94%). Finally, after incorporating the FDL module, LAC rises to 93.28% (+5.39%), while Ave increases to 97.66% (+1.79%).

The ablation study demonstrates that the SCMI module has a significant positive impact on the experimental results, particularly in the Indian Pines and Salinas datasets. The SPE module is identified as a critical factor in significantly improving accuracy across all datasets. The FDL module shows outstanding performance in the Salinas dataset. In summary, the ablation study validates that our proposed SCMI, SPE, and FDL modules significantly improve the performance of the framework.

E. Experimental Results

The experimental results of our proposed HyperSC on the Indian Pines, Houston, and Salinas datasets are presented in Tables III–V. The experimental results are the average values after five runs with different random seeds to ensure stability and reliability.

As the number of classes expands from 6 to 16 across Tasks 1–3, the OA on the Indian Pines dataset decreases from 98.38% to 87.62%. The performance remains relatively

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TABLE IV

CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH CLASS ON THE HOUSTON DATASET

Class No.	Class Name	Task1 (5 classes)	Task2 (5 classes)	Task3 (5 classes)
0	Healthy Grass	81.73±0.29	82.01±0.34	82.34±0.42
1	Stressed Grass	97.91±0.43	98.20±0.64	97.03±1.27
2	Synthetic Grass	98.93±1.15	96.67±1.75	96.79±1.12
3	Tree	98.94±1.05	98.16±1.66	96.00±2.79
4	Soil	100.00±0.00	98.28±2.03	96.44±2.66
5	Water		94.69±0.56	86.43±7.15
6	Residential		90.07±2.57	72.39±11.07
7	Commercial		70.75±5.74	59.28±12.16
8	Road		84.70±5.95	60.17±8.12
9	Highway		88.86±13.77	55.12±16.86
10	Railway			81.95±16.17
11	Parking Lot1			86.34±12.64
12	Parking Lot2			75.79±8.60
13	Tennis Court			99.51±0.97
14	Running Track			99.87±0.17
OA		95.12±0.31	89.42±1.37	80.75±1.07
AA		95.50±0.36	90.24±1.24	83.03±0.81
Kappa		93.81±0.39	88.10±1.54	79.11±1.15

TABLE V

CLASSIFICATION RESULTS (%) OF OUR METHOD FOR EACH CLASS ON THE SALINAS DATASET

Class No.	Class Name	Task1 (6 classes)	Task2 (6 classes)	Task3 (4 classes)
0	Brocoli Green Weeds1	100.00±0.00	100.00±0.00	100.00±0.00
1	Brocoli Green Weeds2	100.00±0.00	100.00±0.00	99.86±0.19
2	Fallow	100.00±0.00	100.00±0.00	99.91±0.13
3	Fallow Rough Plow	100.00±0.00	99.51±0.10	99.10±0.44
4	Fallow Smooth	99.73±0.07	98.74±0.22	97.77±0.50
5	Stubble	100.00±0.00	100.00±0.00	100.00±0.00
6	Celery		99.91±0.02	99.63±0.28
7	Grapes Untrained		99.59±0.14	74.92±4.01
8	Soil Vinyard Develop		100.00±0.00	100.00±0.00
9	Corn Senesced Green Weeds		99.30±0.11	95.63±0.77
10	Lettuce Romaine 4wk		99.82±0.09	99.08±0.36
11	Lettuce Romaine 5wk		99.99±0.02	99.98±0.03
12	Lettuce Romaine 6wk			100.00±0.00
13	Lettuce Romaine 7wk			99.79±0.09
14	Vinyard Untrained			94.15±1.86
15	Vinyard Vertical Trellis			99.85±0.06
OA		99.95±0.01	99.73±0.03	93.28±0.74
AA		99.95±0.01	99.74±0.02	97.48±0.27
Kappa		99.94±0.02	99.69±0.04	92.51±0.82

stable between Tasks 1 and 2 (OA: 98.38% → 96.26%), but a significant drop of 8.64% occurs between Tasks 2 and 3. This decline primarily stems from the spectral overlap between newly introduced crop categories (e.g., Corn Notill and Soybean Notill) and existing classes. For example, the accuracy of Corn Mintill drops from 99.52% in Task 1 to 84.69% in Task 3. Notably, infrequent classes such as Alfalfa achieve an accuracy of 94.36%, demonstrating the effectiveness of our method in handling class-imbalanced learning. However, fine-grained distinctions among soybean cultivation types remain challenging.

The Houston dataset highlights the challenges of urban scenes, where OA declines from 95.12% (Task 1) to 80.75% (Task 3). Artificial structures such as commercial and highways suffer significant performance degradation due to spatial texture similarity. In contrast, structured targets like tennis courts and running tracks maintain near-perfect accuracy (>99%) owing to their distinct geometric patterns. This underscores our method's strong capability in recognizing regular spatial features but reveals limitations in disentangling complex urban textures.

In the Salinas dataset, although OA declines to 93.28% in Task 3, the AA remains robust at 97.48%, indicating stable performance across most classes. Categories with distinct spectral signatures, such as broccoli green weeds1 and stubble, achieve 100% accuracy across all tasks, attributed to their unique reflectance profiles. However, grapes untrained exhibits a drastic accuracy drop from 99.59% in Task 2 to 74.92% in Task 3, primarily due to the limited coverage of auxiliary samples across its growth stages.

The experiments demonstrate that our method excels in spectrally dominant scenarios (Salinas), but faces challenges in spatial–spectral coupled urban environments (Houston) and fine-grained agricultural classification (Indian Pines).

F. Comparative Experiments

To comprehensively evaluate the performance of our proposed HyperSC method, we have selected some conventional exemplar-free lifelong learning methods for comparison, including LWF [26], SDC [30], FeTrIL [46], ADC [31], and FCS [32]. A brief introduction to the comparative methods is provided below.

1) LWF: It uses knowledge distillation to mimic the previous model's outputs and retain old knowledge during new task learning.

2) SDC: The method estimates and compensates for semantic drift by approximating prior task drift through current task data.

3) FeTrIL: It integrates a fixed feature extractor and a pseudo-features generator to enhance stability plasticity.

4) ADC: The method generates adversarial samples by perturbing new samples near old prototypes, and the new model is trained with these samples to estimate drift and update old prototypes.

5) FCS: This method uses a feature calibration network to adapt old-class prototypes via optimal transport and a prototype-involved contrastive loss to separate old and new features, reducing boundary distortion.

We conduct comparative experiments on the Indian Pines, Houston, and Salinas datasets. We focus on the LAC, that is, the accuracy of Task 3 and the Ave. The experimental results are the average values after five runs with different random seeds. Detailed results are presented in Table VI.

On the Indian Pines dataset, our method achieves optimal performance in Tasks 2 and 3 with OAs of 96.26% and 87.62%, respectively, demonstrating absolute improvements of 3.68% and 6.19% over the suboptimal method FeTrIL (92.58% and 81.43%). The average task accuracy (Ave) of our method (94.08%) is 3.29% higher than FeTrIL (90.79%), validating the consistent superiority of our approach in lifelong learning. Notably, LWF and SDC achieve OAs of around 65% in Task 3, highlighting their failure to retain previously learned knowledge when acquiring new tasks.

For the Houston dataset, while FCS marginally outperforms our method in Task 1 with an OA of 96.17%, our approach dominates incremental learning stages: it attains 89.42% OA in Task 2 (a 7.12% lead over FeTrIL's 82.30%) and 80.75% OA in Task 3 (a 15.20% lead over the suboptimal FCS at 65.55%). Our method's average task accuracy (88.43%) surpasses FeTrIL (80.62%) by 7.81%, primarily due to its effective retention of previously learned knowledge.

On the Salinas dataset, our method achieves state-of-the-art OAs of 99.73% (Task 2) and 93.28% (Task 3), with an average task accuracy (97.66%). FeTrIL, as the runner-up, attains OAs

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TABLE VI
COMPARATIVE EXPERIMENTAL RESULTS (%) ON THE INDIAN PINES, HOUSTON, AND SALINAS DATASETS

Dataset	Stage	LWF [26]	SDC [30]	FeTrIL [46]	ADC [31]	FCS [32]	Ours
Indian Pines	Task1	98.38±0.20	97.20±2.46	98.36±0.26	**98.40±0.21**	97.11±0.65	98.38±0.20
	Task2	81.34±3.61	63.57±10.44	92.58±1.91	84.00±3.63	87.53±2.98	**96.26±0.83**
	Task3	65.38±1.72	65.91±2.42	81.43±1.36	73.80±3.16	70.08±1.61	**87.62±1.34**
	Ave	81.70±1.34	75.56±3.95	90.79±0.92	85.40±1.96	84.91±1.30	**94.08±0.66**
Houston	Task1	95.12±0.31	95.23±0.32	95.37±0.13	95.23±0.32	**96.17±0.32**	95.12±0.31
	Task2	48.09±2.84	46.26±4.57	82.30±1.86	48.81±7.36	76.84±0.58	**89.42±1.37**
	Task3	40.17±3.17	41.71±1.68	64.20±1.70	38.99±5.10	65.55±2.31	**80.75±1.07**
	Ave	61.12±1.46	61.07±1.90	80.62±0.99	61.01±3.05	79.52±0.89	**88.43±0.80**
Salinas	Task1	99.95±0.01	99.96±0.01	**99.97±0.00**	99.96±0.01	99.89±0.03	99.95±0.01
	Task2	71.58±5.11	71.32±5.36	98.11±0.41	83.04±4.99	96.24±0.77	**99.73±0.03**
	Task3	42.30±6.73	43.89±6.25	90.32±0.43	49.05±5.58	86.67±0.64	**93.28±0.74**
	Ave	71.28±3.32	71.72±3.63	96.13±0.25	77.35±3.34	94.27±0.43	**97.66±0.25**

TABLE VII
COMPARATIVE CLASSIFICATION RESULTS (%) OF THE LAST TASK ON THE INDIAN PINES DATASET

Class No.	Class Name	LWF [26]	SDC [30]	FeTrIL [46]	ADC [31]	FCS [32]	Ours
0	Corn Mintill	50.46±15.34	0.66±0.73	84.16±1.79	32.83±10.04	41.28±17.72	**84.69±2.55**
1	Corn	0.00±0.00	1.09±1.68	**98.26±1.05**	80.54±6.82	97.39±2.34	97.17±0.63
2	Grass Pasture	29.98±10.53	17.09±9.61	69.53±2.97	17.85±12.22	57.81±5.22	**78.30±2.90**
3	Grass Tree	93.00±2.73	86.03±4.63	94.81±1.61	**97.10±1.49**	94.66±2.37	96.79±1.49
4	Hay Windrowed	97.40±3.16	3.64±4.10	99.23±0.27	95.08±9.84	96.17±2.76	**99.86±0.11**
5	Soybean clean	36.95±7.26	12.94±7.81	**81.13±4.51**	44.15±19.81	61.88±3.83	75.71±5.23
6	Wheat	47.90±34.81	76.17±21.58	99.63±0.49	99.63±0.49	97.16±3.23	**100.00±0.00**
7	Build Grass Tree Drives	73.58±6.58	57.33±6.34	77.82±3.40	61.94±7.90	80.30±6.52	**88.18±2.72**
8	Stone Steel Towers	94.22±6.38	94.67±10.67	99.56±0.89	99.56±0.89	98.67±1.09	**100.00±0.00**
9	Alfalfa	64.10±31.28	**96.41±1.26**	94.36±3.77	93.85±4.76	91.28±6.20	94.36±1.92
10	Grass Pasture Mowed	92.73±8.91	98.18±3.64	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
11	Oats	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	96.00±8.00	**100.00±0.00**
12	Corn Notill	78.38±4.48	**90.62±3.86**	74.78±2.85	78.41±6.62	73.29±4.23	90.16±2.26
13	Soybean Notill	66.62±6.01	86.30±2.54	78.43±1.22	82.46±4.41	58.74±4.82	**87.58±3.52**
14	Soybean Mintill	58.77±7.72	**83.86±4.07**	70.24±4.31	76.87±5.80	60.90±5.10	80.53±3.60
15	Woods	79.45±5.48	89.98±5.38	**96.48±0.76**	91.96±3.33	83.70±2.55	95.72±0.89
	OA	65.38±1.72	65.91±2.42	81.43±1.36	73.80±3.16	70.08±1.61	**87.62±1.34**
	AA	66.47±1.72	62.18±2.18	88.65±0.79	78.26±3.13	80.58±1.47	**91.82±0.73**
	Kappa	60.76±1.76	60.43±2.65	78.92±1.49	70.09±3.52	66.07±1.74	**85.84±1.51**

TABLE VIII
COMPARATIVE CLASSIFICATION RESULTS (%) OF THE LAST TASK ON THE HOUSTON DATASET

Class No.	Class Name	LWF [26]	SDC [30]	FeTrIL [46]	ADC [31]	FCS [32]	Ours
0	Healthy Grass	0.00±0.00	12.78±24.40	80.80±0.42	32.82±40.20	**82.77±3.68**	82.34±0.42
1	Stressed Grass	0.00±0.00	0.00±0.00	76.43±9.86	11.17±22.33	96.26±4.46	**97.03±1.27**
2	Synthetic Grass	39.29±42.13	0.00±0.00	95.33±2.50	0.00±0.00	85.82±8.35	**96.79±1.12**
3	Tree	0.00±0.00	0.55±1.10	83.43±6.63	23.16±34.97	**96.00±3.66**	**96.00±2.79**
4	Soil	70.44±25.74	0.32±0.48	85.93±3.58	5.28±6.51	**99.73±0.40**	96.44±2.66
5	Water	0.00±0.00	38.46±32.01	73.43±4.87	56.78±16.14	71.33±4.13	**86.43±7.15**
6	Residential	71.85±13.80	**85.50±4.15**	56.88±3.09	62.95±15.42	62.57±22.85	72.39±11.07
7	Commercial	13.79±4.36	17.28±7.98	30.96±8.44	17.61±9.55	33.54±5.67	**59.28±12.16**
8	Road	59.17±15.13	**74.90±9.47**	55.22±4.37	12.69±9.87	48.44±10.35	60.17±8.12
9	Highway	2.72±3.09	29.29±0.90	26.83±0.68	3.17±6.33	18.46±8.47	**55.12±16.86**
10	Railway	81.35±4.36	89.07±6.88	71.40±5.32	**93.95±4.21**	56.43±9.89	81.95±16.17
11	Parking Lot1	63.86±19.85	78.37±12.34	35.16±5.07	**92.80±0.96**	33.39±6.00	86.34±12.64
12	Parking Lot2	68.42±2.72	**76.98±3.14**	67.93±2.20	72.91±5.41	44.35±18.49	75.79±8.60
13	Tennis Court	97.98±0.57	**99.84±0.32**	98.70±1.39	99.43±0.61	92.63±6.73	99.51±0.97
14	Running Track	90.40±6.11	**100.00±0.00**	92.35±2.97	**100.00±0.00**	99.70±0.29	99.87±0.17
	OA	40.17±3.17	41.71±1.68	64.20±1.70	38.99±5.10	65.55±2.31	**80.75±1.07**
	AA	43.95±3.38	46.89±1.72	68.72±1.41	45.65±4.34	68.09±2.01	**83.03±0.81**
	Kappa	35.41±3.49	37.27±1.90	61.44±1.82	34.48±5.49	62.68±2.46	**79.11±1.15**

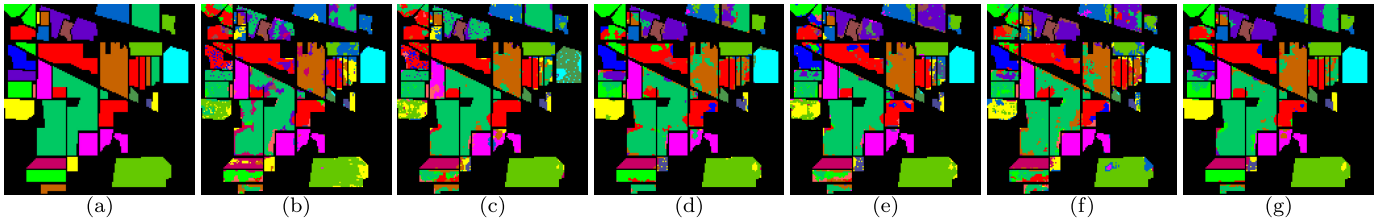
of 98.11% (Task 2) and 90.32% (Task 3), lagging behind our method by 1.62% and 2.96%, respectively. LWF, SDC, and ADC perform poorly in Task 3 with OAs between 40 and 50%, underscoring their inability to mitigate catastrophic forgetting.

Tables VII–IX present classification results of all compared methods across all learned classes after completing the last task, Task 3. In Table VII, our method achieves the best classification performance in nine classes in the Indian Pines dataset,

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TABLE IX
COMPARATIVE CLASSIFICATION RESULTS (%) OF THE LAST TASK ON THE SALINAS DATASET

Class No.	Class Name	LWF [26]	SDC [30]	FeTrIL [46]	ADC [31]	FCS [32]	Ours
0	Brocoli Green Weeds1	0.00±0.00	0.00±0.00	99.99±0.02	59.10±46.11	99.73±0.33	**100.00±0.00**
1	Brocoli Green Weeds2	1.13±2.24	0.03±0.07	98.33±0.62	10.75±21.11	91.56±6.79	**99.86±0.19**
2	Fallow	11.25±11.08	0.41±0.81	76.51±9.75	10.50±17.04	89.59±10.40	**99.91±0.13**
3	Fallow Rough Plow	0.22±0.44	0.00±0.00	**99.98±0.03**	41.76±47.40	99.95±0.07	99.10±0.44
4	Fallow Smooth	1.32±2.33	1.08±2.01	98.74±0.25	0.34±0.26	**99.26±0.62**	97.77±0.50
5	Stubble	0.00±0.00	39.03±47.79	99.55±0.56	60.21±46.31	95.95±4.95	**100.00±0.00**
6	Celery	88.23±13.53	79.02±18.56	97.80±2.72	93.50±8.89	98.23±1.88	**99.63±0.28**
7	Grapes Untrained	13.56±6.39	0.00±0.00	**76.40±3.97**	5.13±4.38	75.71±3.81	74.92±4.01
8	Soil Vinyard Develop	65.79±37.88	80.96±20.87	99.80±0.10	55.14±33.57	96.11±4.44	**100.00±0.00**
9	Corn Senesced Green Weeds	72.81±16.91	69.42±20.84	85.76±6.59	73.99±19.81	71.14±8.45	**95.63±0.77**
10	Lettuce Romaine 4wk	70.53±17.52	89.49±5.58	96.43±3.12	72.47±15.00	**99.52±0.63**	99.08±0.36
11	Lettuce Romaine 5wk	0.00±0.00	8.48±16.06	98.40±1.26	26.11±37.96	88.82±4.94	**99.98±0.03**
12	Lettuce Romaine 6wk	97.77±4.40	**100.00±0.00**	99.58±0.34	**100.00±0.00**	99.39±0.23	**100.00±0.00**
13	Lettuce Romaine 7wk	99.75±0.22	**99.89±0.13**	99.31±0.48	99.36±0.42	98.02±0.74	99.79±0.09
14	Vinyard Untrained	96.34±2.58	**99.99±0.02**	83.36±3.08	99.33±0.77	73.53±1.82	94.15±1.86
15	Vinyard Vertical Trellis	99.88±0.14	**99.98±0.05**	98.46±1.05	98.39±1.54	89.46±5.51	99.85±0.06
	OA	42.30±6.73	43.89±6.25	90.32±0.43	49.05±5.58	86.67±0.64	**93.28±0.74**
	AA	44.91±4.65	47.98±4.89	94.27±0.78	56.63±4.44	91.62±1.01	**97.48±0.27**
	Kappa	36.99±6.65	38.68±6.62	89.18±0.46	44.35±5.78	85.11±0.71	**92.51±0.82**



(a) (b) (c) (d) (e) (f) (g)

Fig. 7. Classification maps for the last task (Task 3) in the Indian Pines dataset. (a) Ground truth. (b) LWF. (c) SDC. (d) FeTrIL. (e) ADC. (f) FCS. (g) Ours.

while also maintaining a fairly balanced performance across other classes. In terms of overall classification performance, our method ranks first with OA = 87.62%, AA = 91.82%, and Kappa = 85.84%. Similarly, as shown in Table VIII, our method achieves the best classification results in six classes in the Houston dataset, leading in OA = 80.75%, AA = 83.03%, and Kappa = 79.11%. Additionally, as shown in Table IX, our method achieves the best classification results in nine classes in the Salinas dataset. Consistent with the Indian Pines and Houston datasets, our method also achieves the optimal overall classification performance with OA = 93.28%, AA = 97.48%, and Kappa = 92.51%.

Across all tasks, our method exhibits smaller than or equal to standard deviations compared to other methods in the majority of stages, confirming its exceptional robustness during training.

These comprehensive experimental results demonstrate that HyperSC, our proposed method, attains optimal lifelong learning performance across three datasets. This is primarily attributed to our proposed SCMI module, which maximizes spectral consistency between the generated auxiliary samples and real samples. Additionally, our SPE module effectively incorporates as much information as possible about the data distribution of previous tasks into the auxiliary samples. Consequently, we generate high-quality auxiliary samples for lifelong learning, ensuring the retention of prior knowledge while alleviating catastrophic forgetting. Furthermore, the FDL module in HyperSC enables simultaneous learning of new knowledge and retention of existing knowledge, thereby enhancing overall performance.

G. Visualize Results

Following the quantitative evaluation, we perform a visual analysis between our method and the comparative approaches on the Indian Pines, Houston, and Salinas datasets. Figs. 7–9 illustrate the classification maps for the last task (Task 3).

As shown in Fig. 7 on the Indian Pines dataset, the LWF method introduces excessive red noise in the upper-left quadrant, primarily misclassifying corn (blue) and corn mintill (green) as corn notill (red). Additionally, soybean clean (purple) and grass pasture (yellow) in the upper and left regions are erroneously categorized as soybean mintill (dark green). The SDC method similarly exhibits significant red noise artifacts in the upper-left quadrant, incorrectly classifying corn (blue) and corn mintill (green) as corn notill (red). In the left region, grass pasture (yellow) is misclassified as soybean mintill (dark green), while hay windrowed in the right region is erroneously mapped to soybean mintill (dark green). Both ADC and FCS methods demonstrate misclassification of left-region corn mintill (green) and soybean clean (purple) as soybean mintill (dark green). In the upper-right quadrant, soybean mintill (dark green) is inaccurately classified as soybean clean (purple), accompanied by substantial noise artifacts across multiple regions. Although FeTrIL outperforms the aforementioned methods in visualization quality, it still introduces noticeable

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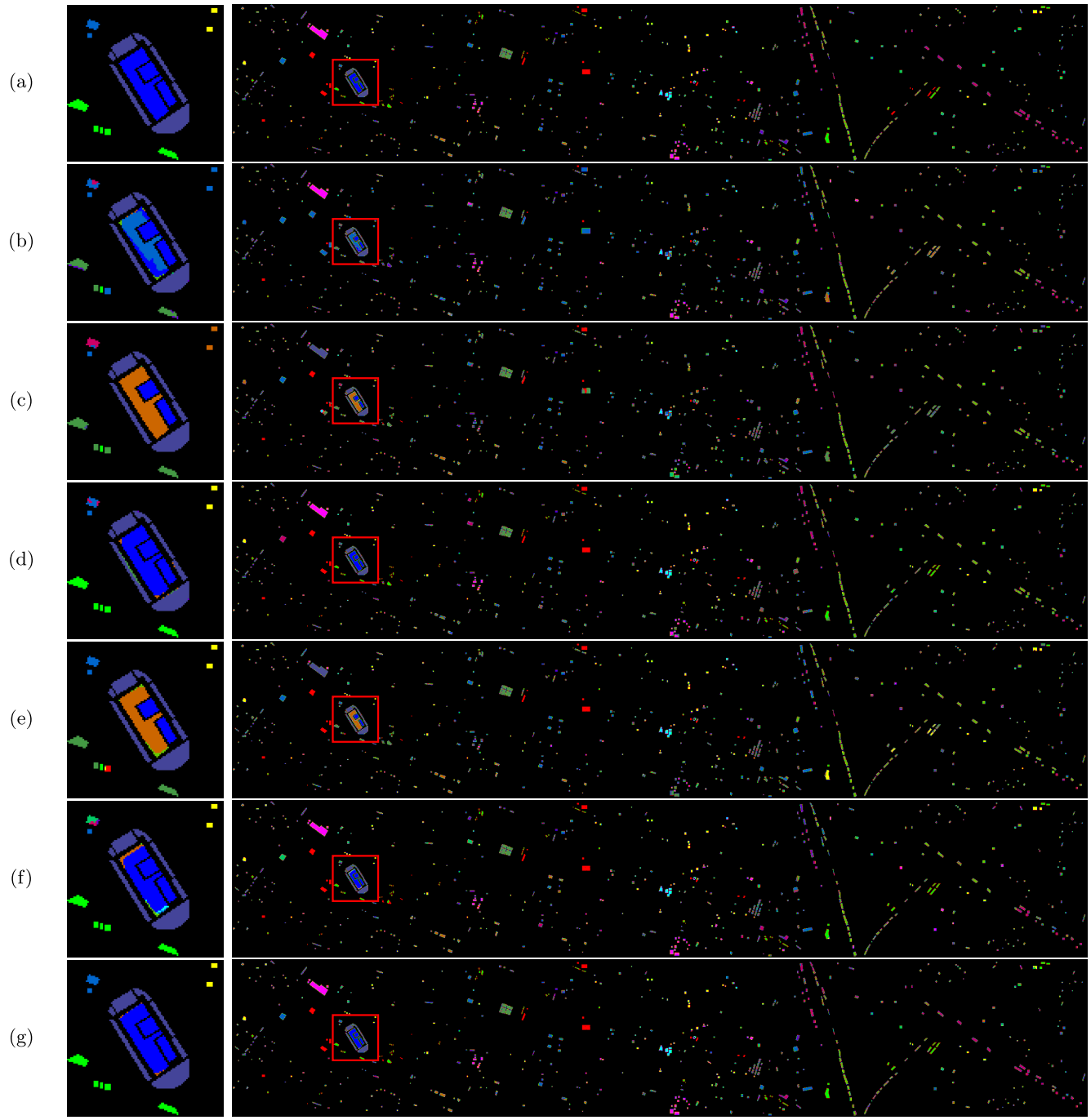


Fig. 8. Classification maps for the last task (Task 3) in the Houston dataset. The left images provide enlarged views of the areas indicated by red boxes in the corresponding right images. (a) Ground truth. (b) LWF. (c) SDC. (d) FeTrIL. (e) ADC. (f) FCS. (g) Ours.

noise across various regions. Our proposed method, while exhibiting minor misclassifications such as mapping soybean mintill (dark green) to soybean clean (purple) in the upper right corner, achieves minimal noise contamination and the closest resemblance to ground-truth label visualization.

In the Houston dataset, as shown in Fig. 8, we employ localized magnification of critical regions for enhanced visualization clarity. The LWF method introduces excessive light blue noise within the playground. Both ADC and SDC introduce too much orange noise in the playground. These three methods primarily misclassify synthetic grass into other categories. The FCS method introduces green noise in the upper-left corner and orange noise along the upper edge of the playground. FeTrIL introduces some red noise in the upper-left corner and orange noise at the edges of the playground. Although our method also introduces a small amount of red noise at the edges of the playground, it maintains significantly clearer boundaries and is very close to the ground truth.

On the Salinas dataset, as shown in Fig. 9, the LWF, ADC, and SDC methods introduce substantial noise in several areas: the deep green region on the left (grapes untrained), the red region (broccoli green weeds1), the green region in the middle

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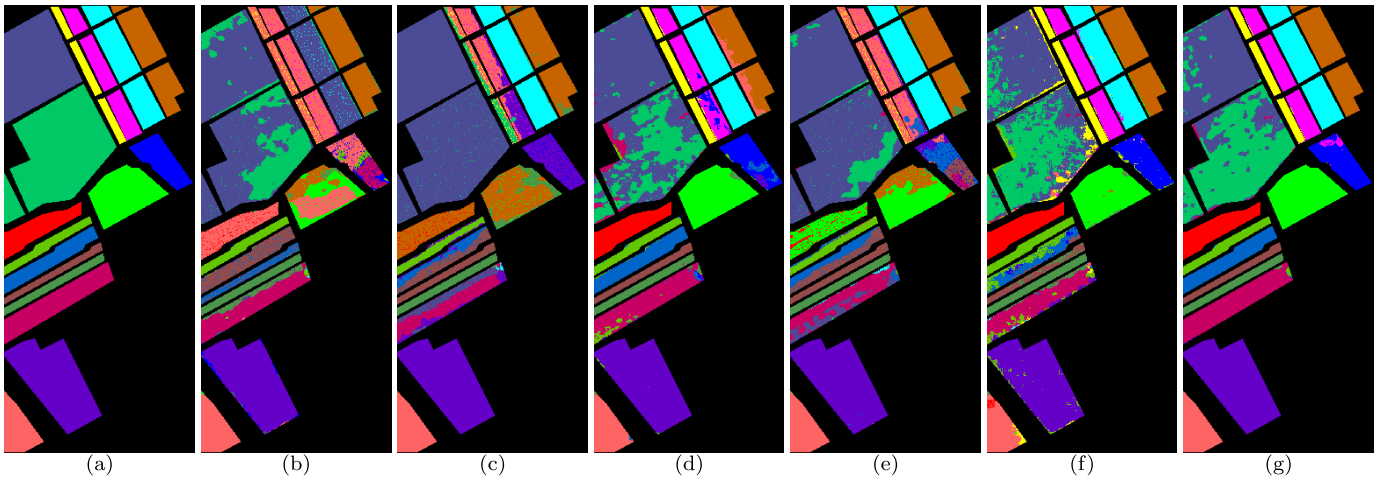


Fig. 9. Classification maps for the last task (Task 3) in the Salinas dataset. (a) Ground truth. (b) LWF. (c) SDC. (d) FeTrIL. (e) ADC. (f) FCS. (g) Ours.

(broccoli green weeds-2), and the light red region at the top (fallow smooth). The FCS method primarily introduces noise in the purple region on the left (vineyard untrained) and the light green region (grapes untrained). FeTrIL mainly adds some noise to the light green region on the left (grapes untrained). Although our method has some noise in the middle light green area (grapes untrained), it overall has the least noise and is closest to the ground truth.

H. Discussion

While the proposed framework, HyperSC, demonstrates competitive performance, it still has certain limitations. First, the model inversion process for auxiliary sample generation introduces nonnegligible computational overhead, even though it produces only a single batch per task. Second, the quality variance of generated samples directly impacts model stability, which reveals an inherent stability-efficiency tradeoff in pseudo-rehearsal approaches. To address these limitations, we will explore leveraging intermediate statistics preserved during prior task training to maintain critical information about historical samples, thereby circumventing both the computational overhead of explicit sample generation and the quality instability inherent in current pseudo-rehearsal paradigms.

V. Conclusion

In this study, we propose HyperSC, a novel exemplar-free lifelong learning framework for HSI classification. HyperSC consists of three key modules: the SCMI module, the SPE module, and the FDL module. The SCMI module addresses the challenge of generating high-quality samples for HSIs due to the high-dimensional spectral bands by minimizing the discrepancy in data distribution between generated and real samples of the corresponding class across all spectral bands. The SPE module injects class-specific Gaussian noise, generated based on the statistics of the original training data, into auxiliary samples across each spectral band, and applies momentum-based updating to enhance sample diversity while preserving spectral fidelity. The FDL module integrates auxiliary and real samples for distillation learning, enabling it to learn the current task while mitigating catastrophic forgetting. Experiments show that our proposed method outperforms previous related methods in exemplar-free lifelong learning.

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